LPG & INDUSTRIAL GASES LOADING ARMS
Series 1802
Technical specifications

Liquid Transfer Srl

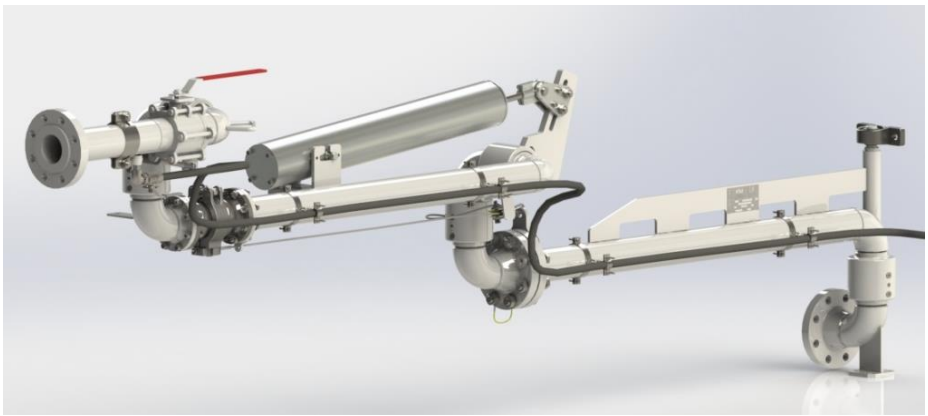
DOC: 1802_01_EN **REV:** 01-09/2017

The series 1802 bottom loading arms are designed to transfer LPG, Industrial Gases or pressurized liquids like Ammonia in closed circit system. The special design allows a wide working area with a compact solution.

Standard configuration

- **Right-hand layout, bottom inlet with connecting flange ANSI 300**
- **Base joint style F-20:** it handles the horizontal movement.
- **Boom pipe**: it is used to estend the range of the loading arm.
- **F-50 style Double swivel joint** : It is used for horizontal and vertical rotations. It's made with two carbon steel swivel joints with double row of ball bearings and NBR seals.
- **Balancing system with compressed spring piston**
- **Secondary Pipe:** high strength carbon steel.
- **ERC Emergency Release Valve (DN50-DN80)**: PN40 in AISI316 is used to avoid leakage of liquid or vapours in case of accidental departure of the truck during loading.
- **F-50 style end double swivel jonts:** this allows vertical and horizontal movement.
- **Ball valve in high strength steel**: carbon steel body, AISI304 Ball, seals in PTFE in ''Fire-safe'' execution.
- **½'' Drainage ball valve**: it is used depressurize the end of the arm between the ball valve and the tank coupler: carbon steel body, AISI304 Ball, seals in PTFE in ''Fire-safe'' execution.
- **Secondary balancing system of the end part with gas piston.**
- **Terminal connection with lap joint flange or ACME threaded coupler or Stanag coupler.**

Technical features				
Nominal diameter		1 ½"	2"	3"
Fluid type		LPG-Hydrocarbons		
Nominal flow rate	m^3/h	18	50	80
	l/min	300	840	1380
Design temperature		-25°C / +65°C		
Weight (kg)		100	120	140
Design pressure		25 bar		
Test pressure		40 bar		
Working pressure		10 bar		

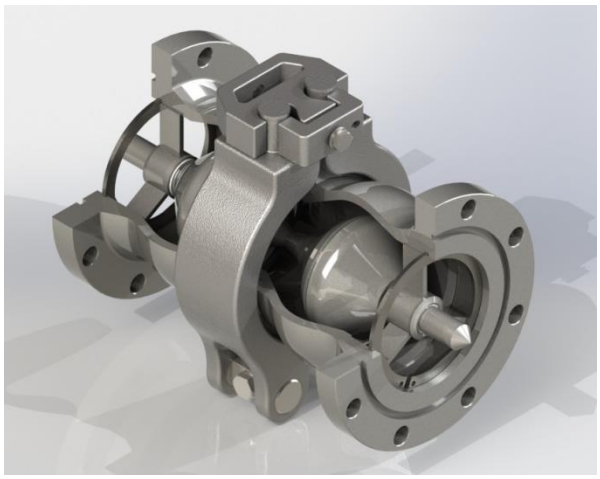


Standards and Regulations

- Conformity Declaration of current Directive **PED** for Pressure Equipment.
- Conformity Declaration of current Directive **ATEX** for Equipment used in Potentially explosive atmospheres.
- Conformity Declaration of current Directive **MACHINERY**
- Customs Declaration of certification for Russia, Kazakhstan, Belarus, **EAC certification**.
- Standard **API-ASTM-ANSI-TTMA.**

Accessories

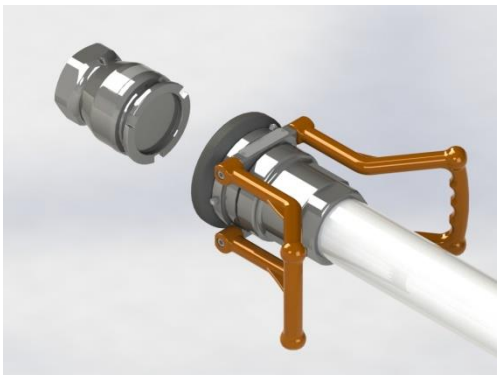
- **Ball valve position Micro-switch**
- **Arm position Micro-switch** (load or park)
- **½'' drainage pipe**
- **Shock absorber for ERC**, to stress gradually the cable and avoid accidental breaking
- **Stand-post**



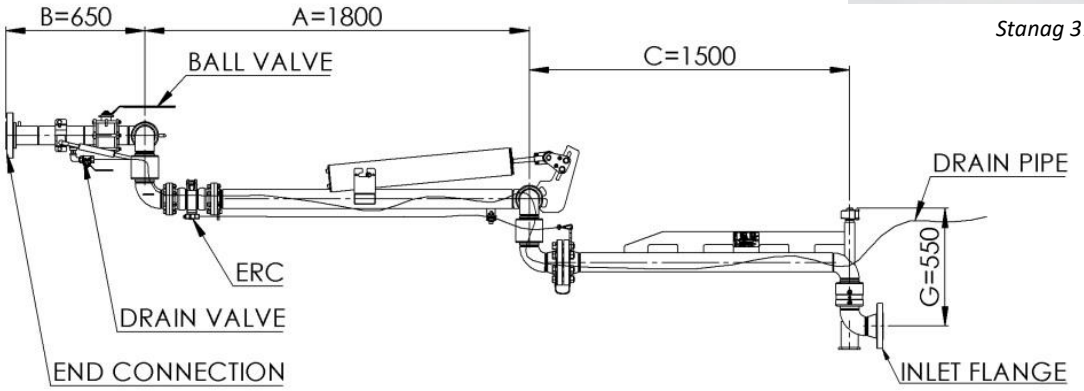
ERC Emergency Release Valve

Options upon request

- **Arm material:** carbon steel for low temperature, stainless steel AISI304 or AISI316 with supply ball valve
- **Seals in HNBR, FKM, FFKM, EPDM, PTFE**
- **Left version**
- **Connecting flange PN40** or other standards
- **Test at 60 bar** for nominal pressure of 40 bar (PN40)
- **Split Type joint**: in 3 pieces for easy maintenance
- **Special configurations** for extreme temperatures (-60/+200 °C)
- **Terminal connections:**
  - **Quick dry-coupler Stanag 3756**
  - **ACME theaded coupler**
  - **NPT thread male or female**



Stanag 3756 dry-coupler



Dimensions in mm

A = 1500-2500 (std 1800)
B = 650-1000 (std 650)
C = 1000-2000 (std 1500)
Dimensions can be customized.

Standard documentation

- Declaration of conformity to regulations
- Declaration of material conformities and functional test (CCC)
- Operation and maintenance manual (MUM)

Documentation on request

- **Welding book (WB):**
  - Welding map (WM)
  - Welding qualification (PQR)
  - Welding specifications (WPS)
  - Welder qualification (WQ)
  - Penetrant liquids test
  - Radiographs of welding heads
- **Materials specifications map (MIM):**
  - Certification 3.1 EN 10204 for steel
  - Certification 2.2 EN 10204 for aluminium
- **Quality complete plan (QCP):**
  - Welding dossier (WB)
  - Materials identification map (MIM)
  - Manufacturing plan

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